



Project Review



View Project

SUPPORT CLUSTER: Mechatronics, Electronics and Digital

EMAIL: FEIT-MEDLabCluster@unimelb.edu.au (<mailto:FEIT-MEDLabCluster@unimelb.edu.au>).

WEB: <https://unimelbcloud.sharepoint.com/sites/FEIT-LabTech-MED>

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ID	2095
Status	APPROVED
Project Title	An IoT System to Improve Overall Equipment Efficiency in Manufacturing
Description	This industry-sponsored capstone project focuses on designing and developing an Internet of Things (IoT)-enabled system to improve Overall Equipment Efficiency (OEE) within the manufacturing operations of Nissan Casting Australia Pty (NCAP). The company currently records shift-level OEE metrics manually, a process that is inefficient, opaque, and prone to human error. To address these limitations, the student team will design and implement a web-based platform for streamlined data entry and automated data collection from selected equipment. The system will include an intuitive front-end interface, robust server-side architecture, and a well-designed database capable of supporting long-term operational use. The end-to-end prototype delivered through this project will demonstrate how digitalisation and IoT concepts can provide actionable insights into manufacturing performance, support continuous improvement, and reduce operational inefficiencies. As part of the collaboration, students will have opportunities and are expected to engage directly with NCAP engineers, participate in monthly site visits to observe real factory environments, capture what forms and information must be collected, develop the application, and execute trials with the system. Students will experience firsthand how digital systems integrate into industrial workflows. This project offers a practical pathway for students to contribute to meaningful real-world challenges while building skills aligned with contemporary industry expectations.
Project Code	EM-12-ATAN-001
Project Type	INDUSTRY
Industry/Organisation	Nissan Casting Australia Pty. Ltd.

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